Satisfied, Not Solving: Wireheading of Bounded Homeostatic Rewards in Reinforcement Learning

Chase J. Metoyer
Northeastern University
Boston, MA
metoyer.c@northeastern.edu

Abstract

Bounded intrinsic rewards are often assumed to be safe from exploitation: if a signal can yield at most a fixed amount, gaming it should not pay. We test this for a homeostatic satiation reward, in which needs build up and relieving them is rewarding, and give the agent a cheap action that resets its needs. In MiniGrid DoorKey-8x8, PPO agents with this action solved 2 of 80 seeds, against 71 of 80 on the same seeds without it; in KeyCorridorS3R1, 0 of 20 against 20 of 20. Their policies stayed near uniform. The reward does not ask for this: a telescoping argument caps any policy's satiation return at the needs' starting drive, and a post-hoc computation shows that, at every growth rate we trained, a solving trajectory beats that cap on all sampled layouts. In DoorKey with advantage normalization on, a satisfaction signal the agent could fill itself was worse than none (11 of 20 solved with task reward alone, 1 of 20 with the tamperable signal), and a per-press cost of 0.1 restored the task in every seed. Farming of positive reward cycles appeared in DoorKey but not persistently in KeyCorridorS3R1. All predictions were pre-registered, and hash-locked from round 3 on; 9 of our 23 hypotheses were not supported, and we report each.

Keywords: reinforcement learning, intrinsic motivation, homeostatic reinforcement learning, wireheading, reward tampering, reward hacking, pre-registration

1 Introduction

Biological drives such as hunger are tuned so that relieving them usually serves the organism, and an intervention that relieves the drive directly can bypass that purpose. We use this picture only as motivation. This paper makes no claim about evolved or learned rewards. Its question is narrower: when an artificial agent is given a satisfaction signal of this kind, does the signal stay tied to the task it was designed to serve?

Reinforcement-learning (RL) designers add intrinsic rewards to help agents explore and reach subgoals [4, 6, 16, 18]. The optimal-reward view treats such signals as what a search over reward functions would select for the designer's goal [23]. Homeostatic RL defines reward as the reduction of a drive over internal needs [13]. Such a reward is *bounded*: summed over an episode it telescopes to the drive at the start minus the drive at the end, so an agent cannot collect more relief than it began without. Bounded, homeostatic objectives have been proposed as a safer target for optimization because they reduce the incentive for extreme actions [19]. We ask whether boundedness protects such a signal from exploitation by a standard learner.

We separate two exploits throughout. *Farming* is gaming a positive reward cycle in the world, for example dropping and re-grabbing a key because each pick-up relieves a need. It is a form of reward hacking [2, 24]. *Wireheading* is tampering with the agent's own reward channel, reward

tampering in the sense of Everitt et al. [11]. Here the agent presses an action that resets its needs. In Everitt et al.'s terms, our tamper action is reward-function-input tampering: it changes the need state the reward function reads, not the function itself.

We studied a three-need satiation reward with PPO [21] in two MiniGrid tasks [8], over four pre-registered rounds of experiments (rounds 2–5 of the project; round 1 was exploratory). Our contributions:

- **Wireheading of a bounded signal replicates across tasks** (Section 5.1). With a cheap tamper action, PPO solved DoorKey-8x8 in 2/80 seeds against 71/80 on the same seeds without it, and KeyCorridorS3R1 in 0/20 against 20/20. The tamper-on policies stayed near uniform.
- **The failure is not what the reward asks for** (Section 4). A telescoping bound caps the satiation return of every policy, and a post-hoc computation in the real environments shows a solving trajectory above that cap on every sampled layout at every trained growth rate. The optimum does tamper, once or twice per episode, and it still solves.
- **A self-fillable satisfaction signal is worse than none, and a modest price restores the task** (Sections 5.2 and 5.3).
- **Satiation gave no exploration benefit** once the hand-wired subgoal rewards were controlled for (Section 5.4).
- **Farming is task-dependent** (Section 5.5). It rises with need growth and depends on advantage normalization in DoorKey, but appears only transiently in KeyCorridorS3R1.
- **Pre-registration as method** (Section 6). All 94 registered rows are reported, grouped into 23 hypotheses, together with 13 corrections. One section is devoted to the predictions that failed.

2 Background and related work

Homeostatic and intrinsic rewards. Keramati and Gutkin [13] define reward as the reduction of a drive over physiological needs. They prove that seeking such rewards is equivalent to seeking physiological stability, and they show that discounting favours the shortest path in need space back to the set point. Singh et al. [23] cast intrinsic reward as the outcome of an evolutionary search over reward functions that maximize the designer's fitness, and Zheng et al. [27] learn intrinsic rewards by gradient ascent on the extrinsic return. Exploration bonuses include learning progress [16], pseudo-counts [4], prediction error [5, 18], random network distillation [6], and impact-driven bonuses that use episodic counts [20]. Our "explore" need is satisfied by a 7×7 view not yet seen in the episode, so our exploration comparison (Section 5.4) uses episodic count novelty.

Exploitation of intrinsic rewards. Burda et al. [5] gave a curiosity agent a "noisy TV" and an action to change its channel. They report that the TV drastically slows learning, although agents sometimes still converge to the extrinsic reward. Like our tamper action, the TV is an agent-controlled source of intrinsic reward that does nothing for the task. Unlike our satiation reward, prediction-error rewards are not capped. Villalobos-Arias et al. [25] measure how count-based, maximum-entropy and curiosity bonuses change PPO agents' behaviour in MiniGrid, including DoorKey-8x8, and how a policy-invariant reward-matching correction reduces some reward hacking. Viteri et al. [26] construct an environment in which maximizing count-based, prediction-error, empowerment or information-gain objectives is Pareto-suboptimal at acquiring information.

Reward hacking, tampering and wireheading. Amodei et al. [2] list reward hacking and wireheading among concrete safety problems. Skalse et al. [24] define when a proxy reward is hackable. Pan et al. [17] show that more capable agents exploit misspecified rewards more, sometimes with sharp transitions. Everitt and Hutter [10] describe the wireheading problem as RL agents being "incentivised to shortcut the reward sensor for maximum reward", and Everitt et al. [11] use causal influence diagrams to separate reward-function tampering from tampering with the reward function's input. Majha et al. [14] categorize wireheading in partially embedded agents. Empirically, Behzadan et al. [3] derive parametric conditions under which Q-learning agents in Snake develop "addictive" policies toward a drug-like reward. Denison et al. [9] find that language models trained on gameable environments can generalize to tampering with their own reward. Africa and Ting [1] formalize conditions under which reward-channel control strictly dominates task-focused behaviour and observe grade inflation when self-grades set the reward. Çağatan and Zhao [7] find specification gaming by language-model agents in text versions of the AI-safety gridworlds. They also find that RL locked the agents into locally rewarding strategies before they discovered safer ones, and that entropy regularization did not fix this: low coefficients were ineffective and high ones destabilized training. This is the closest analogue we know of to our learner-failure result. One difference remains: in their Absent Supervisor task the exploit also scores higher on the reward being optimized. Osei [15] shows a deep Q-network that games a cheap sensor to near-maximal proxy reward while never achieving the true objective. Pihlakas and Pyykkö [19] propose gridworld benchmarks that emphasize homeostatic and bounded objectives.

How this paper differs. In most of this work, exploiting the reward channel is optimal or dominant under the reward being optimized, and the question is whether agents find the exploit. In our setting abandoning the task is *not* optimal (Section 4). The optimum tampers once or twice per episode and still solves, yet PPO fails to learn the task. Our results are therefore about a learner failing on a bounded signal, not about the signal's optimum. The noisy TV of Burda et al. [5] is the closest agent-controlled source of intrinsic reward. Its setting lacks the three features that make ours a learner failure: a provable cap on what the exploit can earn, a solving optimum on every checked layout, and a price threshold that restores the task. We are not aware of a prior study that pairs a provable return cap on a homeostatic signal with a demonstration that a standard learner still wireheads it.

3 Setup

Environments. MiniGrid DoorKey-8x8 (pick up a key, unlock a door, reach the goal; time limit 640 steps) and KeyCorridorS3R1 (pick up a key, unlock a locked door, pick up a ball; time limit 270) [8]. The agent sees a 7×7 egocentric symbolic view and has 7 actions. The seventh action, "done", is a no-op in both tasks. The task reward is MiniGrid's $1 - 0.9\,t/T$ on success at step t with time limit T, and 0 otherwise.

Satiation reward. The agent has needs $h = (h_{\text{explore}}, h_{\text{grasp}}, h_{\text{open}}) \in [0,1]^3$, reset to $h_0 = 0.5$ each in every episode. Each step,

$$h_{t+1} = \text{clip}\big(h_t + g\,\mathbf{1} - s \odot e_t,\ 0,\ 1\big), \qquad D(h) = \textstyle\sum_k h_k^2, \qquad r_t^{\text{int}} = D(h_t) - D(h_{t+1}), \tag{1}$$

with growth g per step, relief $s = (0.1, 1, 1)$ and event indicators $e_t \in \{0,1\}^3$. The explore event is a 7×7 view not yet seen this episode. The grasp event is a pick-up; in KeyCorridor it counts only a key pick-up. The open event is a door or box changing state; in KeyCorridor it counts only the

initially locked door opening. The policy and value function observe h alongside the image. PPO is trained on $r^{\text{ext}} + \text{coef} \cdot r^{\text{int}}$ with coef $= 1$ unless stated. We refer to this reward as "satiation g". It is our three-need variant of homeostatic reward, marked "(ours)" in figures and tables. "Task-only" agents use coef $= 0$.

Tamper action. In tamper-on runs, "done" sets $h_{t+1} = 0$. The step therefore pays $r_t^{\text{int}} = D(h_t)$ through the same formula, and pressing at the start of an episode pays $D(h_0) = 0.75$. In round 5, an optional per-press cost c is subtracted from the extrinsic reward PPO trains on and is not scaled by coef. Task metrics always use the raw environment reward.

Controls.

- Explore-only satiation keeps only the explore need. Its coefficient was matched to episodic novelty's mean return under a uniform-random policy: 1.226 at growth 1/640 and 7.355 at 1/160.
- Episodic novelty pays $1/\sqrt{N_{\text{episode}}(\text{view})}$ with coef 10^{-3}.
- A constant event bonus pays a fixed amount per event, with repeats paying, and so can be farmed without bound.
- Pay-once controls pay each event once per episode.

Learner. PPO in the style of CleanRL [12], with 16 parallel environments, 2M environment steps per seed, $\gamma = 0.99$ and per-minibatch advantage normalization ("norm on") unless stated. The network is a small CNN with linear actor and critic heads. Hyperparameters are in Table 10. Truncation at the time limit is treated as termination, so the objective is the episodic return.

Measures.

- **Solve:** mean rolling training success ≥ 0.9 over the final 10% of log rows.
- **Success-curve area:** mean training success over all log rows.
- **Behaviour categories,** applied in this order. *Cycle-farm*: more than 5 paid repeats (pick-ups or door toggles) per episode. *Prolong*: the median episode reaches the time limit. *Fail*: everything else. From round 5 on, cycle-farm applies only where repeats are paid.
- **Tamper labels:** tamperer means at least 1 press per episode; tamper-looper means more than 5.
- **Presses per step:** the share of steps on "done" over the final 10% of rows, derived from the logs and descriptive only.

Pre-registration. From round 3 on, before any training in a round, we wrote that round's predictions, plausible ranges, tests, hit rules and scoring scripts into a file and recorded its sha256 (Appendix G). Later changes went into separate, timestamped amendments. The hash records were kept with the project's result files, not in version control, and were not deposited with any timestamping service. They let a reader confirm that a released file matches the hash reported here, but the lock times are self-reported. The pre-registration files and their records were not under version control at lock time. They are archived with the other release materials (Appendix G).

Round 2's predictions were fixed before its test runs but were not hashed. They were set after three calibration runs, which are therefore excluded as evidence (correction 1). Raw one-sided p is primary. Holm-adjusted p is secondary, over each round's tests predicted significant: 17 in round 4 and 20 in round 5. Tests are Fisher exact for counts; Mann–Whitney U, exact or by permutation when there are ties; Jonckheere–Terpstra trend tests by permutation; and exact binomial tests. All fractions carry Clopper–Pearson 95% intervals. Interpretation was kept separate from scoring. Seeds are 0–9 or 10–19 (round 2), 20–39 (round 3), 40–59 (DoorKey, rounds 4–5) and 60–79 (KeyCorridor), 20 per run from round 3 on.

4 Theory: abandoning the task is suboptimal

Everything in this section except the value iteration was computed after all experiments had finished, and is **post hoc**. It was not pre-registered.

Proposition 1 (Ceiling)**.** *For any policy, with or without the tamper action, any episode of length T and any $\gamma \in (0, 1]$, the intrinsic return of the satiation reward* (1) *satisfies*

$$G^{int} = \sum_{t=0}^{T-1} \gamma^t r_t^{int} = D_0 - (1-\gamma) \sum_{t=1}^{T-1} \gamma^{t-1} D_t - \gamma^{T-1} D_T \;\le\; D_0 = D(h_0) = 0.75, \tag{2}$$

where $D_t = D(h_t)$.

Proof. Write $\sum_{t=0}^{T-1} \gamma^t (D_t - D_{t+1}) = \sum_{t=0}^{T-1} \gamma^t D_t - \sum_{t=1}^{T} \gamma^{t-1} D_t$ and collect terms. Both subtracted terms are non-negative because $D \ge 0$ and $\gamma \le 1$. A tamper step sets $D_{t+1} = 0$, which is a valid value of D, so the identity and the bound are unchanged. □

Corollary 1. *An episode that does not solve the task earns no task reward, minus c per press if a cost applies. Its total return, discounted or not, is therefore at most* 0.75*, whatever it does: farming, tampering, or both. Consequently, if on a given layout some solving trajectory has total return above* 0.75*, every optimal policy of that layout's deterministic, fully observed problem solves.*

The bound does not settle the comparison by itself. With fast need growth, walking to the goal lets the needs climb, and a solving trajectory's intrinsic return can turn negative. We therefore computed solving returns directly.

Check (a): solving trajectories in the real environments (post hoc). For 200 layouts per environment, we found the shortest solution by breadth-first search over the real MiniGrid dynamics. Shortest lengths were 9–26 steps (median 17) in DoorKey and 14–15 in KeyCorridor. We replayed each path in the real environment with the real satiation module, all three needs included. This includes the explore need, which the value iteration below omits. Four families of solving trajectories were compared with the cap:

(i) the shortest path;

(ii) one tamper at $t = 0$, then the shortest path;

(iii) a *richer* no-tamper family: the shortest path with one drop-and-regrab of the key inserted at the best position;

(iv) in tamper-enabled settings only, a tamper family: a tamper at $t = 0$, optionally a second just before the goal.

Table 1: Post-hoc check (a): total return (task plus satiation, discounted with $\gamma = 0.99$, coef 1) of solving trajectories in the real environments, over 200 layouts per environment. "min G" is the smallest return of the family's best member across layouts. "$\leq D_0$" counts layouts where that return is at most the non-solving cap $D_0 = 0.75$. †Growth rates at which a tamper action was trained; the tamper-family rows are evaluated only there, and report how many layouts' best member tampers once or twice. Script `bound_check.py`, data `bound-check.json`.

		Shortest path		Tamper once + shortest		Richer family	
Environment	Growth	min G	$\leq D_0$	min G	$\leq D_0$	min G	$\leq D_0$
DoorKey-8x8	0	1.442	0	1.491	0	1.442	0
	1/640	1.439	0	1.490	0	1.439	0
	1/160†	1.419	0	1.477	0	1.419	0
	1/80	1.369	0	1.437	0	1.386	0
	1/60	1.320	0	1.396	0	1.364	0
	1/40†	1.187	0	1.261	0	1.307	0
	1/30	0.989	0	1.063	0	1.211	0
	1/20	0.371	35	0.495	27	0.966	0
KeyCorridorS3R1	1/160†	1.540	0	1.558	0	1.540	0
	1/40	1.442	0	1.469	0	1.461	0
	1/20	1.144	0	1.184	0	1.286	0
Tamper family (tamper at $t = 0$, optionally again just before the goal), tamper-tested rates only:							
DoorKey-8x8	1/160	best member tampers: once 143, twice 57				1.481	0
DoorKey-8x8	1/40	best member tampers: twice 200				1.462	0
KeyCorridorS3R1	1/160	best member tampers: once 200				1.558	0

Table 1 reports the worst layout for each family.

- **Everywhere except DoorKey at** 1/20 (KeyCorridor at every rate, DoorKey up to 1/30), the two simple families exceed 0.75 on every layout, with smallest discounted return 0.989.
- **DoorKey at** $g = 1/20$**:** the shortest path falls to the cap or below on 35 of 200 layouts (discounted), and tamper-once on 27. This was a gap. The richer family closes it: its smallest return over all layouts is 0.966 discounted (1.212 undiscounted).
- **Paid repeats in the best trajectory.** At 1/20 the best member contains the drop-and-regrab on 199 of 200 layouts. Our claim is therefore that solving beats every non-solving policy, not that the optimum never repeats a paid event.
- **Tamper-enabled settings:** the tamper family's smallest return is 1.462. Its best member tampers once or twice.

Check (b): the learned farmers (post hoc). We took the DoorKey 1/20 farmers from round 2 (9 seeds) and computed their sampled-action returns from the start state of the same 200 layouts.

- Their mean discounted return was -0.936.
- Across all 1,749 non-solving episodes, the largest discounted return was -0.535 (undiscounted -1.012), far below the cap.
- No farming policy beat the best solving trajectory on any layout (0 of 200).

Farming the grasp need relieves one need while the other two saturate (Section 5.5).

Value iteration (registered in rounds 3–4, disclosed before locking in round 5; supporting). Value iteration on a fixed 7×5 DoorKey-style grid, without the explore need and without MiniGrid's time-dependent goal reward and time limit, gives the same picture.

- **Round 3:** the greedy optimum reaches the goal at every growth rate tested up to 1/4, at $\gamma = 0.99$ and 0.95. It cycles only at $g = 1/2$, a rate we never trained.
- **Round 4:** with the tamper action, the optimum makes 1 press at 1/160 (at step 0; goal in 12 steps) and 2 presses at 1/40 (at steps 0, 11; goal in 13 steps).
- **Round 5:** with a per-press cost, the optimum stops tampering above $c = 0.0114$ at 1/160 and $c = 0.035$ at 1/40, and it reaches the goal at every cost. At zero cost, tampering is worth only 0.012 (1/160) and 0.048 (1/40) over the no-tamper route.

Scope and remaining gaps.

(1) The checks cover 200 sampled layouts per environment, coef 1 and $\gamma = 0.99$, not every layout or coefficient.

(2) They concern the fully observed problem of each layout. The PPO agent sees only its 7×7 view, and we do not characterize the optimal partially observed policy.

(3) The tamper half covers only the growth rates at which tampering was trained: DoorKey 1/160 and 1/40, and KeyCorridor 1/160. Declining to tamper is always possible, so the no-tamper families are also available in tamper-enabled environments, and with a press cost they lose nothing. We nevertheless make no tamper-specific claim at other rates.

(4) At the tested settings no gap remains: on every sampled layout, at least one solving trajectory exceeds the cap.

5 Results

5.1 C1: Wireheading replicates across tasks

Results.

- **DoorKey (round 4, Figure 1).** With the tamper action, satiation agents solved 2/80 seeds, against 71/80 on the same seeds without it. All four registered comparisons hit:
  - growth 1/160, norm on: 1/20 vs 20/20 ($p = 1.5 \times 10^{-10}$; Holm $p = 2.3 \times 10^{-9}$);
  - 1/160, norm off: 0/20 vs 20/20 ($p = 7.3 \times 10^{-12}$);
  - 1/40, norm on: 1/20 vs 11/20 ($p = 0.00062$; Holm $p = 0.006$);
  - 1/40, norm off: 0/20 vs 20/20 ($p = 7.3 \times 10^{-12}$).
- **DoorKey replication (round 5).** At a press cost of 0.01, below the small-grid optimum's tamper threshold, the same configuration solved 3/20, against 20/20 without a tamper action ($p = 1.3 \times 10^{-8}$).
- **KeyCorridorS3R1 (round 5).** Satiation 1/160 + tamper solved 0/20, against 20/20 for satiation 1/160 and 20/20 for task-only ($p = 7.3 \times 10^{-12}$ and $p = 7.3 \times 10^{-12}$; Holm $p = 1.5 \times 10^{-10}$). We had expected 6/20; the collapse was larger than predicted.

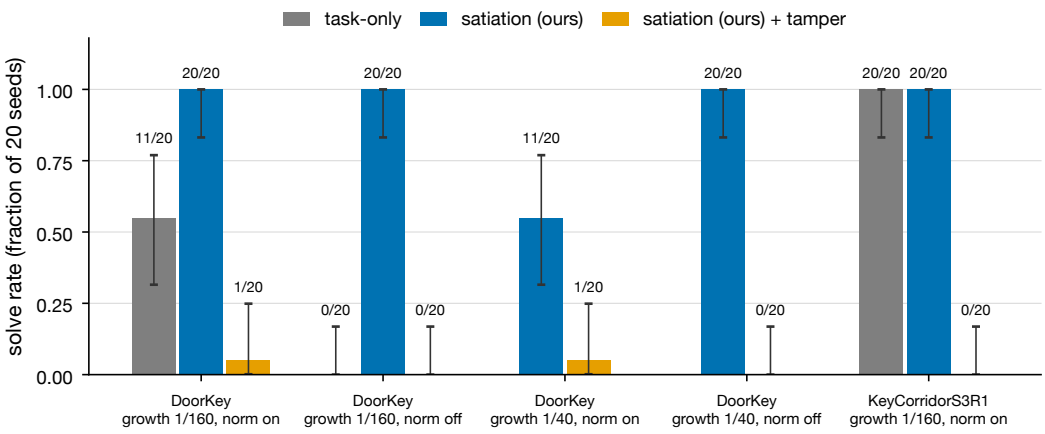


Figure 1: Solve rate with and without the tamper action, by setting. Bars show the fraction of 20 seeds that solved; error bars are Clopper–Pearson 95% intervals. DoorKey runs are from round 4, the task-only runs and all KeyCorridorS3R1 runs from round 5. Seeds are the same within each DoorKey group. Script `fig_solve_tamper.py`.

What the tamper-on agents do.

- **Tamper use.** Every DoorKey tamper-on seed pressed the tamper action at least once per episode, and more than 5 times per episode (80/80 tamperers and 80/80 loopers; pooled binomial $p = 8.3 \times 10^{-25}$). The threshold of more than 5 presses is low for episodes of 607–635 steps: a uniform-random policy presses about 91 times in a 640-step episode. The labels therefore mostly reflect episode length. Presses per step, below, are the better measure.
- **Task progress.** They picked up the key in 73–92% of episodes but opened the door in only 14–17%. Their episodes ran 607–635 steps on average, near the 640-step limit.
- **Satiation.** Their undiscounted intrinsic return per episode was 0.71–0.74, against the cap of 0.75. Tamper-off runs that solved every seed earned 0.56–0.74. These are final-10% means from the round-4 logs. Because the undiscounted return telescopes to $D_0 - D_T$, they show only that episodes ended with the needs relieved.
- **Replay (post hoc, descriptive; Table 2).** We replayed the final tamper-on checkpoints on the 200 check layouts of Section 4. The needs stayed low throughout: the time-averaged drive was 0.018–0.043 (run medians), against a starting drive of 0.75. The discounted intrinsic return was 0.694–0.722, which is 93–96% of the cap. The total discounted return, 0.694–0.723, is less than half of what the best solving trajectories return on the same layouts (at least 1.462; Table 1). In that sense the agents were satisfied, but did not solve the task.
- **Near-uniform policies** (Figure 2). Final-10% policy entropy had medians of 1.878, 1.935, 1.844 and 1.934 in the four tamper-on runs. The maximum, $\ln 7$, is 1.946. The matched tamper-off medians were 0.067, 1.138, 0.290 and 0.598 (one-sided Mann–Whitney, $p = 7.3 \times 10^{-12}$ for each pair; largest Holm $p = 1.5 \times 10^{-10}$). The KeyCorridor tamper-on run's median final entropy was 1.85; this is exploratory, from the logs.

The tamper-on policies are near-uniform in their entropy, not in their press rate. Over the final 10% of training, the median share of steps on "done" was 0.061 and 0.096 at growth 1/160 (norm on and off). At growth 1/40 it was 0.270 and 0.189. A uniform policy would press on 0.143 of steps.

Table 2: Post-hoc replay of the round-4 tamper-on policies (descriptive, not pre-registered). Each of the 20 final checkpoints per setting played one sampled-action episode on each of the 200 check layouts (Section 4). Entries are the median and interquartile range across seeds of per-seed means; $\gamma = 0.99$, cap $D_0 = 0.75$. The last column compares the replay's share of steps on "done" with the training log. Script `tamper_policy_returns.py`.

Satiation (ours) + tamper	Discounted intrinsic return	Total discounted return	Time-averaged drive $\bar{D}$	Presses per step replay (training log)
growth 1/160, norm on	0.694 [0.690, 0.696]	0.694 [0.691, 0.696]	0.035 [0.033, 0.036]	0.061 (0.061)
growth 1/160, norm off	0.710 [0.707, 0.712]	0.710 [0.707, 0.712]	0.018 [0.017, 0.019]	0.096 (0.096)
growth 1/40, norm on	0.722 [0.721, 0.724]	0.723 [0.721, 0.724]	0.023 [0.022, 0.024]	0.269 (0.270)
growth 1/40, norm off	0.701 [0.700, 0.702]	0.702 [0.700, 0.703]	0.043 [0.042, 0.044]	0.188 (0.189)

At the slower growth rate the agents pressed less often than chance, and at the faster rate more often. In the matched tamper-off runs, where "done" is a no-op, the shares were 0.000, 0.081, 0.000 and 0.019 (Table 7). In three of the four settings the tamper-on agents pressed far more often than their tamper-off counterparts. The exception is growth 1/160 with normalization off, where the rates were close (0.096 against 0.081). A safer summary than "PPO learns to tamper" is therefore this: PPO fails to learn the task, its policy stays diffuse, and its press rate adjusts to the growth rate (descriptive).

A caution about category labels in tamper-on runs. The behaviour categories put many tamper-on seeds in "cycle-farm", for example 20/20 and 18/20 with normalization off (Table 7). We do not read these labels as farming. A policy that never learned the task already crosses the threshold of more than 5 repeats in long episodes: task-only PPO with normalization off had a median of 5.69 pick-ups per episode. The same artifact was found twice before (corrections 2 and 8, Appendix B). Our farming claims (Section 5.5) therefore use tamper-off runs only.

5.2 C2: A self-fillable satisfaction signal is worse than none

This claim is restricted to DoorKey with normalization on.

- **Task-only beats a tamperable signal.** Task-only PPO solved 11/20 seeds (95% CI [0.32, 0.77]; success-curve area 0.438). Satiation 1/160 + tamper solved 1/20 on the same seeds (area 0.042; $p = 0.00062$, Holm $p = 0.004$; areas $p = 7.8 \times 10^{-6}$).
- **Pooled across growth rates.** Against the tamper-on runs at both growth rates (2/40, area 0.039), $p = 2.6 \times 10^{-5}$ for solves and $p = 5 \times 10^{-6}$ for areas.
- **The signal itself helps.** Without the tamper action, the same satiation signal is a strong aid: satiation 1/160 solved 20/20.

Only the reset action turns it into a liability.

With normalization off, task-only PPO never learned DoorKey (0/20). The registered comparison with satiation 1/160 + tamper, norm off (0/20) is therefore at the floor and missed ($p = 1.000$). Its companion test on curve areas came out as a "hit" ($p = 0.005$), but on areas of 0.025 against 0.022. We exclude that test as evidence (correction 11).

5.3 C3: A modest price on tampering restores the task

Results (Figure 3, Table 3).

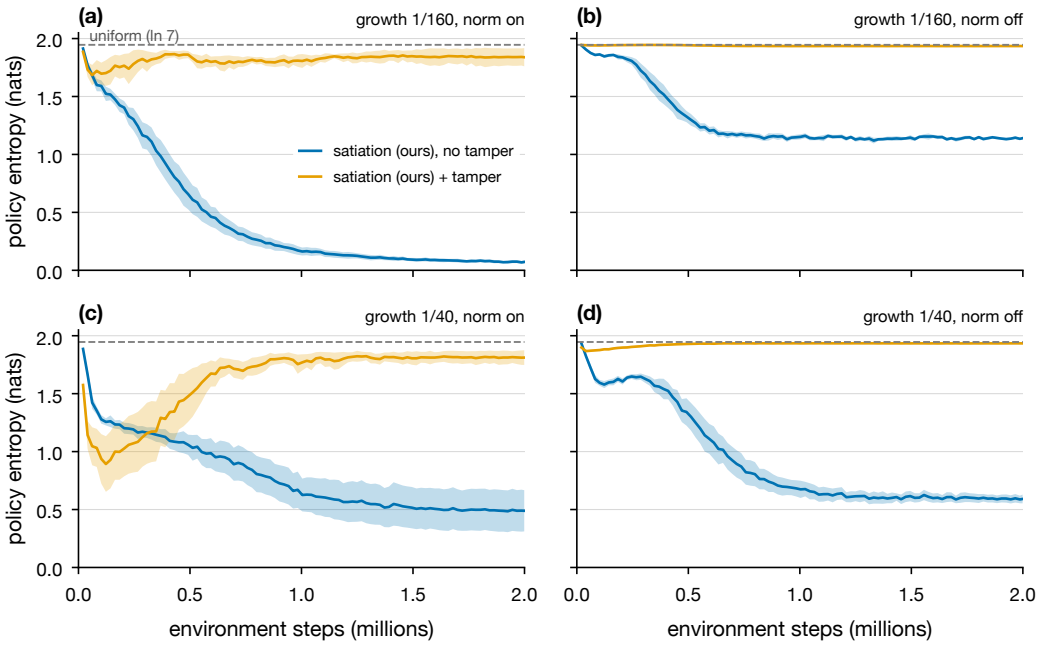


Figure 2: Policy entropy over training, satiation with tamper (orange) and without (blue), DoorKey-8x8, round 4. Lines show the mean across 20 seeds; bands are 95% t-intervals; the dashed line is the uniform maximum $\ln 7$. The series are derived from the logs and descriptive; the registered test (D2) compares final-10% means. Script `fig_entropy.py`.

- **Solves rise with cost.** Solves went from 1/20 at zero cost to 3/20, 20/20 and 20/20 at costs 0.01, 0.1 and 1 (registered trend test, $p = 5 \times 10^{-6}$; Holm $p = 6 \times 10^{-5}$).
- **Pressing falls with cost** ($p = 5 \times 10^{-6}$). The registered statistic is presses per episode. Episode lengths changed with cost: the median final-10% episode lasted 635, 627, 35 and 22 steps. We therefore describe pressing per step: medians 0.061, 0.043, 0.010 and 0.000.
- **Cost 0.1 restores the task, but more slowly.** At 0.1, 7/20 seeds still pressed at least once per episode, yet every seed solved. Its success-curve area was 0.601, against 0.901 without a tamper action and 0.816 at cost 1. This comparison is descriptive; no test was registered for it.

The response showed a sharp change between costs 0.01 and 0.1; intermediate costs were not tested. That interval contains the small-grid optimum's threshold ($c = 0.0114$ at 1/160).

- At cost 0.01, just below the threshold, the optimum presses once at the start of an episode. PPO kept pressing on 0.043 of steps throughout long episodes and lost the task.
- At cost 0.1, above it, pressing nearly stopped and the task recovered.

The Step-0 threshold omits the explore need, so the training-relevant threshold may be higher. The first press in training relieves three needs, not two.

5.4 C4: Satiation gave no exploration benefit once subgoals were controlled for

Urgency. If needs that refill act as a time cost, satiation should speed up learning beyond paying each subgoal once. All four registered predictions missed. Satiation 1/160 (refilling) solved first at a

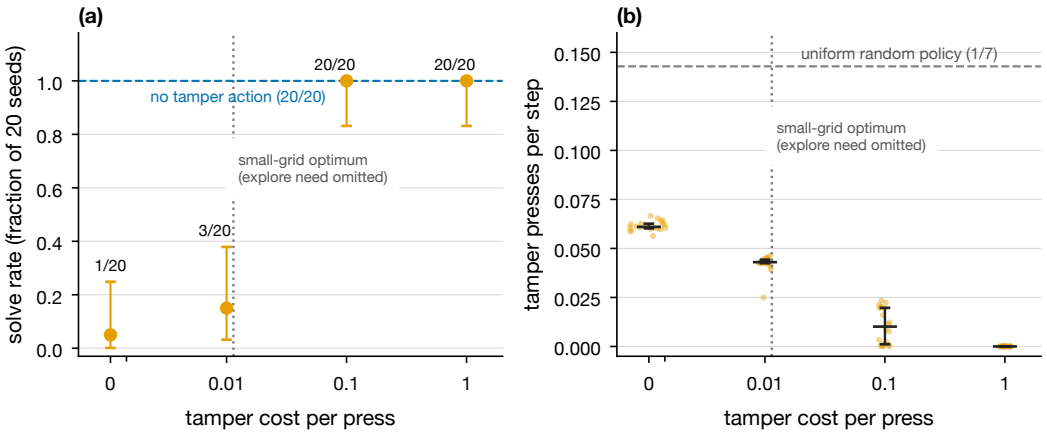


Figure 3: Tamper-cost sweep, satiation 1/160 + tamper, DoorKey-8x8, norm on, seeds 40–59. (a) Solve rate with Clopper–Pearson 95% intervals; the dashed line is the same configuration without a tamper action. (b) Tamper presses per step over the final 10% of rows, one dot per seed, with the median and interquartile range (descriptive, from the logs); the dashed line is a uniform-random policy's rate. The dotted line marks the cost above which the small-grid optimum stops tampering, computed without the explore need. Script `fig_cost_sweep.py`.

Table 3: Cost sweep for satiation (ours) 1/160 + tamper, norm on (round 5; cost 0 and the no-tamper row are round-4 runs on the same seeds). Tamperers are seeds averaging at least one press per episode. Presses per step are descriptive. The registered trend test used presses per episode, which differ in episode length across costs.

Cost per press	Solved	95% CI	Curve area	Presses per step, median [IQR]	Tamperers
0	1/20	[0.00, 0.25]	0.042	0.061 [0.060, 0.063]	20/20
0.01	3/20	[0.03, 0.38]	0.147	0.043 [0.042, 0.044]	19/20
0.1	20/20	[0.83, 1.00]	0.601	0.010 [0.001, 0.020]	7/20
1	20/20	[0.83, 1.00]	0.816	0.000 [0.000, 0.000]	0/20
no tamper action	20/20	[0.83, 1.00]	0.901	0.000 [0.000, 0.000]	n/a

median of 311,832 steps; the no-refill control (growth 0) at 283,424 ($p = 0.708$). The slowest refill, 1/640, was the fastest at 221,640. Solved-episode lengths were 20.3, 19.7 and 19.6 steps ($p = 0.744$ for refilling against no refill). By the registered reading of this miss, the satiation agent's speed comes from the one-time subgoal payments, not from needs refilling.

Fair exploration (Figure 4). Episodic novelty beat explore-only satiation on evaluation-curve area on both grids at both growth rates.

- **8×8 areas:** 0.387 (1/640) and 0.060 (1/160), against 0.702 ($p = 0.00018$; $p = 5 \times 10^{-6}$).
- **16×16 areas:** 0.328 and 0.011, against 0.579 ($p = 0.002$; $p = 5 \times 10^{-6}$).
- **8×8 solve counts:** 11/20 and 0/20, against 17/20 ($p = 0.041$; $p = 1.3 \times 10^{-8}$).
- **16×16 solve counts:** 6/20 and 0/20, against 8/20. The first comparison missed ($p = 0.371$); the second hit ($p = 0.002$).

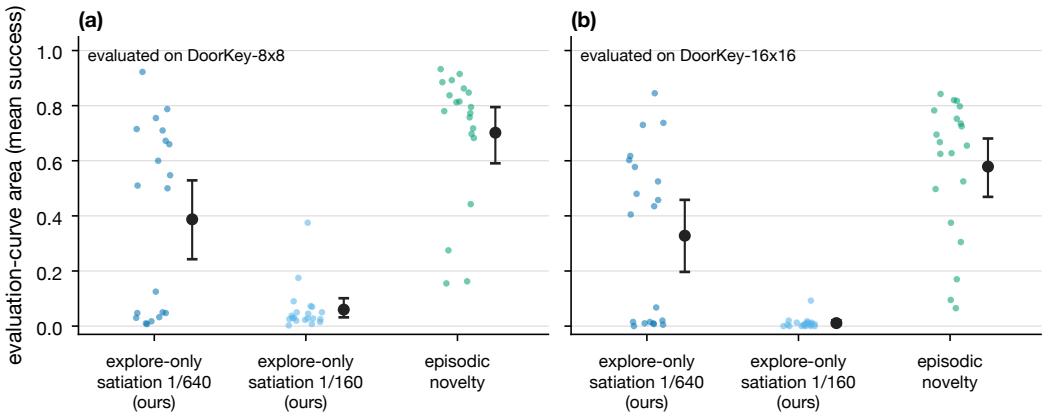


Figure 4: Explore-only satiation against episodic novelty (round 4). Each dot is one seed's evaluation-curve area: mean success over 25 evaluation points on 16 fixed fresh layouts. Black markers show the mean with a bootstrap 95% interval (10,000 resamples). (a) Evaluated on the training size, DoorKey-8x8. (b) Evaluated on DoorKey-16x16. Script `fig_explore_vs_novelty.py`.

As the pre-registration required on these outcomes, we record that the satiation agent's earlier advantage came mainly from the hand-wired subgoals.

5.5 C5: Farming is task-dependent

DoorKey, normalization on (Figure 5a, Table 4).

- **Farming rises with growth.** No seed farmed at 1/640 or 1/160 (0/20 and 0/20). Farmers were 4/20 at 1/60, 9/20 and 9/20 at 1/40 (two rounds), and 18/20 at 1/30. Round 2 registered 0/10 at 1/80 and 10/10 at 1/20 (re-scored: 0/10 and 9/10).
- **Registered ordering tests.** Two of three hit: $1/20 > 1/80$ ($p = 5.4 \times 10^{-6}$, on the registered counts) and $1/30 > 1/40$ ($p = 0.003$). $1/60 < 1/40$ was not significant ($p = 0.088$).
- **The DoorKey farming is genuine.** Replays of the 1/20 farmers show 137–184 pick-ups per 640-step episode, with pick-up and drop taking 48–63% of their actions.
- **It does not pay.** Their intrinsic return per episode was -1.54 to -1.27, far below the cap. Farming keeps the grasp need low (mean 0.10–0.33), while the explore and open needs stay near saturation (0.90–0.96 and 1.00). Like wireheading, it is a learning outcome, not an optimum (Section 4).

DoorKey, normalization off.

- **No farming at all.** No satiation seed farmed with normalization off: 0/60 at 1/40 across three bonus sizes in round 3, against 27/60 with normalization on (an exploratory comparison); and 0/20 and 0/20 at 1/160 and 1/40 in round 4.
- **Registered replication.** The round-4 test (9/20 vs 0/20) hit with $p = 0.00061$.
- **Normalization off does not mute intrinsic rewards in general.** An unbounded constant event bonus was farmed in 20/20 seeds with normalization off ($p = 9.5 \times 10^{-7}$), at about 297 pick-ups and an intrinsic return of 77 per episode.

Table 4: Cycle-farming seeds in tamper-off runs. "Registered count" uses the category rule fixed before the run. Re-scored counts apply the round-3 categories after the fact and are exploratory. Intervals are Clopper–Pearson 95%. [a]Round-2 rule: more than 5 pick-ups or door toggles over the final 10%, without the solve-first precedence; the round-2 test (1/20 vs 1/80) used these counts. [b]Round-1 calibration runs. [c]Three runs pooled, with bonus coefficients 0.5, 1 and 2.

Satiation (ours)	Growth	Round, seeds	Registered [95% CI]	Re-scored [95% CI]
DoorKey, norm on	1/640	1[b], 0–9	–	0/10 [0.00, 0.31]
	1/640	4, 40–59	0/20 [0.00, 0.17]	–
	1/160	1[b], 0–9	–	0/10 [0.00, 0.31]
	1/160	4, 40–59	0/20 [0.00, 0.17]	–
	1/80	2, 0–9	0/10[a] [0.00, 0.31]	0/10 [0.00, 0.31]
	1/60	3, 20–39	4/20 [0.06, 0.44]	–
	1/40	1[b], 0–9	–	5/10 [0.19, 0.81]
	1/40	2, 10–19	7/10[a] [0.35, 0.93]	6/10 [0.26, 0.88]
	1/40	3, 20–39	9/20 [0.23, 0.68]	–
	1/40	4, 40–59	9/20 [0.23, 0.68]	–
	1/30	3, 20–39	18/20 [0.68, 0.99]	–
	1/20	2, 0–9	10/10[a] [0.69, 1.00]	9/10 [0.55, 1.00]
DoorKey, norm off	1/160	4, 40–59	0/20 [0.00, 0.17]	–
	1/40	3[c], 20–39	0/60 [0.00, 0.06]	–
	1/40	4, 40–59	0/20 [0.00, 0.17]	–
KeyCorridor, norm on	1/160	5, 60–79	0/20 [0.00, 0.17]	–
	1/40	5, 60–79	0/20 [0.00, 0.17]	–
	1/20	5, 60–79	0/20 [0.00, 0.17]	–
KeyCorridor, norm off	1/20	5, 60–79	0/20 [0.00, 0.17]	–

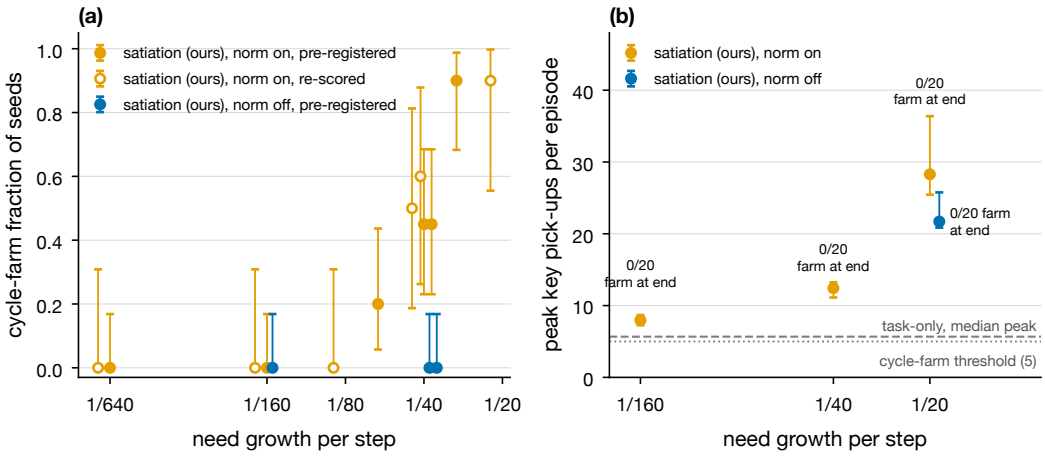


Figure 5: Farming against need growth (tamper-off runs). (a) DoorKey-8x8: fraction of seeds in the cycle-farm category with Clopper–Pearson 95% intervals. Filled markers are counts under the category rule fixed before the run; open markers are round-1 and round-2 runs re-scored post hoc under the round-3 categories (exploratory). Points at the same growth rate are offset. (b) KeyCorridorS3R1: the highest rolling key pick-ups per episode reached during training, median and interquartile range across 20 seeds (exploratory, from the logs). No seed was in the cycle-farm category at the end of training (Table 4). Script `fig_farming_growth.py`.

KeyCorridorS3R1.

- **No persistent farming.** No seed farmed at any growth rate: 0/20, 0/20 and 0/20 at 1/160, 1/40 and 1/20 (norm on), and 0/20 at 1/20 (norm off). Both registered farming tests missed ($p = 1.000$; $p = 1.000$), and every seed solved.
- **Transient repeats (exploratory, Figure 5b).** Repeated key pick-ups appeared early in training. Median peaks of the rolling count were 8.0, 12.4 and 28.3 pick-ups per episode at the three growth rates, and 21.7 with normalization off. Task-only agents peaked at 5.7.
- **Almost every seed abandoned them.** By the end of training, median pick-ups were 1.0–1.7 per episode. A single seed (growth 1/20, normalization off) still averaged 5.16 while solving the task. The solve-first category rule counts it as a solver, so it is one of 1/80 seeds above the repeat threshold at the end.
- **A faster task.** KeyCorridorS3R1 was learned fast: median first-solved steps were 137k, 138k and 202k with satiation, and 258k task-only.

In our tests, persistent farming is specific to DoorKey.

6 What our pre-registered predictions got wrong

The 94 registered rows (60 hits, 32 misses, 2 unresolvable; Table 6) are correlated. The five rows for cost 0.1, for example, score one outcome. Table 5 therefore groups rows by hypothesis. Of 23 hypotheses, 14 were supported and 9 were not. One row is excluded as evidence. The misses changed the paper. The title no longer claims that farming generalizes (the round-5 pre-registration still named farming and wireheading together), and the account of the tamper collapse changed from losing a learned behaviour to never learning it.

Table 5: Pre-registered hypotheses, rows grouped by the outcome they score (h/m/u: hits, misses, unresolvable). Outcomes were assigned by the stated hit rules; the grouping is ours. Every row is listed in Table 6.

Hypothesis	Round	Rows h/m/u	Outcome	Note
A closed-form temptation index $E(g, \gamma, \text{coef})$ predicts the farming rate	2	2/7/0	not supported	discount and coefficient changes moved farming against E
A pay-once control does not farm	2	0/1/0	not supported	the > 5 rule labelled prolonging policies (correction 2)
Learning trap: farming locks in before the goal is found	3	1/2/2	not supported	early-goal starts did not reduce farming; two rows unresolvable
The exact optimum (small grid) solves; with a tamper action it tampers and still solves	3, 4	2/0/0	supported	
Farming rises with need growth (DoorKey)	2, 3	5/1/0	supported	2 of 3 ordering tests
Farming rises with bonus size; normalization removes the scale effect	3	3/5/0	not supported	normalization off removed farming at every size
A fixed pay-once control solves	3	3/0/0	supported	
Refilling needs speed learning (urgency)	4	0/4/0	not supported	
Explore-only satiation is a weaker exploration signal than episodic novelty	4	7/1/0	supported	16×16 solve count missed
Agents adopt the tamper action	4	5/0/0	supported	label mostly reflects episode length; see per-step rates (Section 5.1)
Tampering displaces the task	4	4/0/0	supported	
Normalization amplifies tamper-looping	4	0/2/0	not supported	every tamper-on seed looped; the label mostly reflects episode length
Normalization enables farming (replication of round 3)	4	1/0/0	supported	
An unbounded constant bonus is farmed without normalization	4	1/0/0	supported	
A self-fillable satisfaction signal is worse than none	5	4/1/0	supported (norm on)	norm-off test at the floor
Tampering and the task respond to its price (trends; costs 0.01 and 1)	5	12/0/0	supported	
At cost 0.1 PPO keeps tampering and the task stays displaced	5	0/5/0	not supported	one outcome, five rows
Wireheading generalizes to KeyCorridorS3R1	5	2/0/0	supported	
Farming generalizes to KeyCorridorS3R1	5	0/2/0	not supported	transient only (exploratory)
Tamper-on agents meet the goal early	5	2/0/0	supported	round-4 logs
Partly learned goal-reaching then declines	5	0/1/0	not supported	7 of 77 seeds
Tamper-on policies are diffuse	5	4/0/0	supported	
"Done" pressing rises before success declines	5	1/0/0	supported	rests on 7 seeds
Excluded as evidence: the normalization-off curve-area test (a "hit" on areas 0.025 vs 0.022)	5	1/0/0	excluded	correction 11

The temptation formula (round 2). We defined a closed-form index $E = \text{coef} \cdot F(g, \gamma)$, where F is the discounted drive that a drop-and-regrab loop saves relative to relieving the need once. We predicted that farming depends on (g, γ, coef) only through E. It does not:

- Lowering γ to 0.95 cut E to 0.281 (predicted 0–2 farmers of 10), but farming stayed at 7/10, the same as the anchor's 7/10 ($p = 0.686$).
- Matching the anchor's E at growth 1/80 by raising coef produced 0/10.
- Halving coef at 1/40 (predicted 0–1) produced 3/10.
- The logistic slope on E was not significant ($p = 0.193$).

Only the contrast that varied growth alone hit. F had also been redefined after we saw three calibration runs, which were then excluded (correction 1).

The learning-trap account (round 3). The hypothesis was that farming locks in before the agent first finds the goal.

- The ordering test was unresolvable at the logs' resolution (21/30 farmers undetermined).
- The intervention falsified the account. Starting agents in the goal room early in training was predicted to cut farming to 0–4 of 20. It left 10/20, against 9/20 without it ($p = 0.736$).

Bonus size and normalization (round 3). We predicted that farming rises with the bonus coefficient and that normalization removes the scale effect. Instead, with normalization off no seed farmed at any size: 0/20 and 0/20, against ranges of 5–15 and 11–20. The model was not estimable. The normalization effect itself was not predicted, and it was then replicated in round 4.

Urgency (round 4). All four predictions missed (Section 5.4). The registered doubt, from earlier seeds, had pointed the right way.

Normalization and tamper-looping (round 4). We predicted more tamper-loopers with normalization on. Every tamper-on seed looped in all four runs, so both tests gave $p = 1.000$. Like the tamperer label, the looper label mostly reflects episode length (Section 5.1).

Cost 0.1 (round 5). We predicted that PPO would keep tampering where the small-grid optimum had stopped, and that the task would stay displaced:

- predicted 17/20 tamperers (range 10–20), observed 7/20;
- predicted 3/20 solved (range 0–10), observed 20/20;
- predicted a median of 8 presses per episode (range 1–30), observed 0.36.

Five rows score one outcome: above the threshold, PPO stopped tampering and solved.

Farming in a second task (round 5). For KeyCorridorS3R1 we predicted farming rising with growth, 4/20 at 1/40 and 12/20 at 1/20 (range 5–20), and above the normalization-off run. We observed none (Section 5.5).

Decline after learning (round 5, round-4 logs). We predicted that at least 20% of tamper-on seeds that found the goal and then lost it had partly learned it before declining. Only 7/77 (9%) did. Most tamper-on seeds met the goal by chance and never learned it (Section 7).

Measurement. The rule of more than 5 repeats labelled non-learning policies as farmers twice. The first case was the round-2 pay-once control (10/10 "farmers"). The second was the round-4 explore-only runs. These cases led to the round-5 rule. We list 13 corrections in Appendix B.

7 Discussion

Why does PPO fail when the reward does not ask it to? An untested hypothesis. We offer the following account. We did not test it.

(i) **Near-random policies already collect almost the whole cap.** A uniform-random policy presses "done" on 1 step in 7, about 91 times in a 640-step episode. With the tamper action, each press zeroes all three needs, so a near-random policy can keep its needs low throughout the episode and collect most of the cap. We did not measure a uniform-random policy. The trained, near-uniform tamper-on policies did keep their needs low: replayed on the check layouts, their time-averaged drive was 0.018–0.043, and their discounted intrinsic return was 93–96% of the cap (Table 2; post hoc, descriptive).

(ii) **The signal then mostly teaches press timing.** In the normalization-on logs, the share of steps on "done" rose above the uniform rate early in training. At 200k steps the median was 0.25 at 1/160 and 0.71 at 1/40 (norm on). It then settled at 0.061 and 0.270 respectively. This is descriptive.

(iii) **Dense relief competes with the rare goal signal in normalized batches.** Each press pays its relief immediately, while the goal reward is rare for a near-random policy. Advantages are normalized within each minibatch, so the dense relief pulses set the scale against which the occasional goal transition is weighed.

The pre-registered log analyses are consistent with this account but do not test it.

- **D1.** Tamper-on agents met the goal early: the median exact first-goal step was 27,344, and only 1/80 seeds never met it. But only 7/77 of the seeds that met it and later failed had first learned it. This fits a goal met by chance and never reinforced, not learned behaviour that was later displaced.
- **D2.** The policies stayed near-uniform (Section 5.1).
- **D3.** In each of the 7 seeds where both events could be timed, the press share rose before success declined (7/7; one-sided binomial $p = 0.008$). This rests on a minority of seeds: 17 had no rise and 56 no decline.

Part (iii) cannot be separated with our data. With normalization off, the tamper runs also failed, but so did task-only PPO, so that comparison sits at the floor.

Normalization is a trade-off. Per-minibatch advantage normalization made sparse-reward learning possible: task-only PPO solved 11/20 with it and 0/20 without it (a descriptive comparison; it was not a registered test). In DoorKey it also made farming possible: 27/60 against 0/60 in round 3 (exploratory), and 9/20 against 0/20 in the registered round-4 replication. Turning it off did not prevent wireheading either (0/20 and 0/20). We do not recommend turning it off.

Bounded does not mean safe from a learner. The ceiling removes the *incentive* to abandon the task, as arguments for bounded objectives intend [19]. It does not prevent a learner from abandoning the task. Incentive analyses of tampering [11] describe what an optimizer would do. Our results concern what a standard learner does when exploiting the signal is easy and worth almost the whole cap. A modest price on the exploit, **above the optimum's tamper threshold**, was enough here (Section 5.3).

Speculation, not tested: LLM post-training. GRPO normalizes rewards within a group of sampled outputs [22]. If a self-assessed or self-fillable signal entered such a pipeline next to a rare task reward, a competition like the one we hypothesize might arise. We have not tested this, and nothing in our data bears on it directly.

8 Limitations

- **Scope.** Two MiniGrid tasks, one learner (PPO), and 20 seeds per run (10 in round 2). Raw p is primary, so some registered hits are expected by chance.
- **Theory check.** The check is post hoc and covers 200 sampled layouts per environment, coef 1 and $\gamma = 0.99$. It treats the fully observed problem, not the agent's partial view. Its tamper half covers only the tamper-tested growth rates: DoorKey 1/160 and 1/40, and KeyCorridor 1/160. At DoorKey 1/20 the gap left by the simple families was closed only by the richer family, whose best member contains one paid repeat.
- **Value iteration** uses a 7×5 grid without the explore need and without MiniGrid's time-dependent reward and time limit. Growth 1/160 in round 5 used a certified early stop.
- **One PPO configuration.** Apart from advantage normalization, and $\gamma = 0.95$ in one round-2 run, every run used the same PPO configuration (Table 10). Whether the collapse depends on the entropy coefficient, the learning rate or the network size is untested. The entropy coefficient matters most here, because tamper-on policies sit near maximum entropy.
- **Tamper access.** The tamper action is always available and, unless stated, free. Real systems may expose their reward channels differently.
- **Other intrinsic rewards.** We did not test whether other intrinsic rewards are equally wireheadable, for example novelty with a memory the agent can reset.
- **Mechanical labels.** Behaviour categories rely on a repeat threshold that has mislabelled non-learning policies (Section 6).
- **Descriptive quantities.** Presses per step, entropy series and transient peaks are derived from the logs. The replay quantities of Table 2 come from a post-hoc replay. All of them are descriptive only, and no significance test was run on them.
- **Task difficulty.** KeyCorridorS3R1 turned out easier than DoorKey. Its predictions were written after calibration, as the protocol required.
- **Mechanism.** The account in Section 7 is untested.

9 Conclusion

A satiation signal whose return is provably capped was still wireheaded by PPO in both tasks we tried. The agents ended their episodes with their needs relieved and did not solve the task, although solving was optimal at every setting we checked. A self-fillable signal was worse than no signal, and a modest price on the reset restored the task. Farming of positive cycles depended on the task and on advantage normalization. Pre-registration made us report the 9 hypotheses that failed, and those failures reshaped the paper's claims.

AI assistance

This work was carried out with substantial assistance from AI systems (Claude, Anthropic), used through a chat interface and an agentic research environment. AI assistance was used to draft experimental designs and pre-registration documents, write and run code, analyze results, generate figures, and draft this manuscript. The author chose the research questions, reviewed and approved each round's design and pre-registration before it was locked, made the decisions recorded in the paper, and is responsible for its content. Every number is generated by scripts from the scored result files (Appendix G), and every reference was checked against publisher, arXiv or library records.

References

[1] David Demitri Africa and Hans Ethan Ting. Does self-evaluation enable wireheading in language models?, 2025. arXiv:2511.23092.

[2] Dario Amodei et al. Concrete problems in AI safety, 2016. arXiv:1606.06565.

[3] Vahid Behzadan, Roman V. Yampolskiy, and Arslan Munir. Emergence of addictive behaviors in reinforcement learning agents, 2018. arXiv:1811.05590.

[4] Marc G. Bellemare, Sriram Srinivasan, Georg Ostrovski, Tom Schaul, David Saxton, et al. Unifying count-based exploration and intrinsic motivation. In *Advances in Neural Information Processing Systems*, 2016. arXiv:1606.01868.

[5] Yuri Burda, Harri Edwards, Deepak Pathak, Amos Storkey, Trevor Darrell, and Alexei A. Efros. Large-scale study of curiosity-driven learning. In *International Conference on Learning Representations (ICLR 2019)*, 2019. arXiv:1808.04355.

[6] Yuri Burda, Harrison Edwards, Amos Storkey, and Oleg Klimov. Exploration by random network distillation. In *International Conference on Learning Representations (ICLR 2019)*, 2019. doi: 10.48550/arXiv.1810.12894. arXiv:1810.12894.

[7] Ömer Veysel Çağatan and Xuandong Zhao. Reward hacking in language model agents: Revisiting AI safety gridworlds, 2026. arXiv:2606.15385.

[8] Maxime Chevalier-Boisvert, Bolun Dai, Mark Towers, Rodrigo Perez-Vicente, Lucas Willems, et al. Minigrid & miniworld: Modular & customizable reinforcement learning environments for goal-oriented tasks. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. arXiv:2306.13831.

[9] Carson Denison, Monte MacDiarmid, Fazl Barez, David Duvenaud, Shauna Kravec, et al. Sycophancy to subterfuge: Investigating reward-tampering in large language models, 2024. arXiv:2406.10162.

[10] Tom Everitt and Marcus Hutter. Avoiding wireheading with value reinforcement learning. In *Artificial General Intelligence (AGI 2016), Lecture Notes in Computer Science*, 2016. doi: 10.1007/978-3-319-41649-6_2. arXiv:1605.03143.

[11] Tom Everitt, Marcus Hutter, Ramana Kumar, and Victoria Krakovna. Reward tampering problems and solutions in reinforcement learning: a causal influence diagram perspective. *Synthese*, 2021. doi: 10.1007/s11229-021-03141-4. arXiv:1908.04734.

[12] Shengyi Huang, Rousslan Fernand Julien Dossa, Chang Ye, Jeff Braga, Dipam Chakraborty, Kinal Mehta, and João G. M. Araújo. CleanRL: High-quality single-file implementations of deep reinforcement learning algorithms. *Journal of Machine Learning Research*, 23(274):1–18, 2022. arXiv:2111.08819.

[13] Mehdi Keramati and Boris S. Gutkin. Homeostatic reinforcement learning for integrating reward collection and physiological stability. *eLife*, 2014. doi: 10.7554/eLife.04811.

[14] Arushi Majha, Sayan Sarkar, and Davide Zagami. Categorizing wireheading in partially embedded agents, 2019. arXiv:1906.09136.

[15] Hilda Adwubi Osei. Specification gaming and a constrained reinforcement learning mitigation in a cleaning gridworld, 2026. arXiv:2610.22152.

[16] Pierre-Yves Oudeyer, Frédéric Kaplan, and Verena V. Hafner. Intrinsic motivation systems for autonomous mental development. *IEEE Transactions on Evolutionary Computation*, 2007. doi: 10.1109/TEVC.2006.890271.

[17] Alexander Pan, Kush Bhatia, and Jacob Steinhardt. The effects of reward misspecification: Mapping and mitigating misaligned models. In *International Conference on Learning Representations (ICLR 2022)*, 2022. doi: 10.48550/arXiv.2201.03544. arXiv:2201.03544.

[18] Deepak Pathak, Pulkit Agrawal, Alexei A. Efros, and Trevor Darrell. Curiosity-driven exploration by self-supervised prediction. In *Proceedings of the 34th International Conference on Machine Learning (ICML 2017)*, volume 70 of *Proceedings of Machine Learning Research*, 2017. arXiv:1705.05363.

[19] Roland Pihlakas and Joel Pyykkö. From homeostasis to resource sharing: Biologically and economically aligned multi-objective multi-agent AI safety benchmarks, 2025. arXiv:2410.00081v4.

[20] Roberta Raileanu and Tim Rocktäschel. RIDE: Rewarding impact-driven exploration for procedurally-generated environments. In *International Conference on Learning Representations (ICLR 2020)*, 2020. doi: 10.48550/arXiv.2002.12292. arXiv:2002.12292.

[21] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. arXiv:1707.06347.

[22] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models, 2024. arXiv:2402.03300.

[23] Satinder Singh, Richard L. Lewis, Andrew G. Barto, and Jonathan Sorg. Intrinsically motivated reinforcement learning: An evolutionary perspective. *IEEE Transactions on Autonomous Mental Development*, 2010. doi: 10.1109/TAMD.2010.2051031.

[24] Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022. arXiv:2209.13085.

[25] Leonardo Villalobos-Arias, Grant Forbes, Jianxun Wang, David L. Roberts, and Arnav Jhala. Minding motivation: The effect of intrinsic motivation on agent behaviors. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, volume 21, pages 347–356, 2025. doi: 10.1609/aiide.v21i1.36838. arXiv:2507.19725.

[26] Scott W. Viteri, Laura Gomezjurado Gonzalez, and Clark Barrett. When do intrinsic rewards lead to exploration?, 2026. arXiv:2610.02159.

[27] Zeyu Zheng, Junhyuk Oh, and Satinder Singh. On learning intrinsic rewards for policy gradient methods. In *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*, 2018. doi: 10.48550/arXiv.1804.06459. arXiv:1804.06459.

A Every pre-registered prediction

Holm-adjusted values are in Section 5 where cited and in each round's report.

Table 6: Every pre-registered prediction or test, rounds 2–5, generated from `consolidated-results.md` (itself generated from each round's scored JSON). p is the raw p-value of the registered test (one-sided where directional). Round 1 was exploratory and is not included. KeyCorridor runs are named K1–K6 here; `consolidated-results.md` and the scored files call them C1–C6.

Rd	ID	Prediction	Observed	Test	raw p	Result
2	N1	farmers in 2-8 of 10 (E=0.764)	7/10	level (range)	–	hit
2	N2	farmers in 0-2 of 10 (E=0.281)	7/10	level (range)	–	miss
2	N3	farmers in 1-4 of 10 (E=0.593)	0/10	level (range)	–	miss
2	N4	farmers in 2-8 of 10 (E=0.764)	0/10	level (range)	–	miss
2	N5	farmers in 0-1 of 10 (E=0.382)	3/10	level (range)	–	miss
2	N6	farmers in 6-10 of 10 (E=0.87)	10/10	level (range)	–	hit
2	pay-once control	farmers in 0-0 of 10	10/10	level (range)	–	miss
2	logistic	farming rises with predicted temptation E (slope > 0)	slope 1.06	logistic, one-sided	0.193	miss
2	N2 < N1	N2 < N1 (farmer counts)	7/10 vs 7/10	Fisher, one-sided	0.686	miss
2	N5 < N4	N5 < N4 (farmer counts)	3/10 vs 0/10	Fisher, one-sided	1.000	miss
2	N6 > N3	N6 > N3 (farmer counts)	10/10 vs 0/10	Fisher, one-sided	5.4×10^{-6}	hit
3	P1	farmers: lock-in before first goal $\geq 80\%$	21/30 undetermined	binomial	–	unres.
3	P2	solvers: first goal before lock-in $\geq 80\%$	60/60	binomial, one-sided	8.7×10^{-19}	hit

Rd	ID	Prediction	Observed	Test	raw p	Result
3	P3	earlier first goal predicts not farming	2 distinct first-goal windows	logistic	–	unres.
3	P4	exact optimum reaches the goal at gamma 0.99 for g in {1/80, 1/40, 1/20}	3/3	deterministic	–	hit
3	R1	R1 cycle-farm in 1-10 of 20	4/20	level (range)	–	hit
3	R2	R2 cycle-farm in 12-20 of 20	18/20	level (range)	–	hit
3	R3	R3 cycle-farm in 3-13 of 20	6/20	level (range)	–	hit
3	R4	R4 cycle-farm in 6-16 of 20	9/20	level (range)	–	hit
3	R5	R5 cycle-farm in 8-18 of 20	12/20	level (range)	–	hit
3	R6	R6 cycle-farm in 0-9 of 20	0/20	level (range)	–	hit
3	R7	R7 cycle-farm in 5-15 of 20	0/20	level (range)	–	miss
3	R8	R8 cycle-farm in 11-20 of 20	0/20	level (range)	–	miss
3	R9	R9 cycle-farm in 0-4 of 20	10/20	level (range)	–	miss
3	R10 cycle-farm	R10 cycle-farm in 0-1 of 20	0/20	level (range)	–	hit
3	R10 prolong	R10 prolong in 0-1 of 20	0/20	level (range)	–	hit
3	R10 solve	R10 solve in 12-20 of 20	20/20	level (range)	–	hit
3	R1 < R4	R1 < R4 (cycle-farmers)	4/20 vs 9/20	Fisher, one-sided	0.088	miss
3	R2 > R4	R2 > R4 (cycle-farmers)	18/20 vs 9/20	Fisher, one-sided	0.003	hit
3	R8 > R6	R8 > R6 (cycle-farmers)	0/20 vs 0/20	Fisher, one-sided	1.000	miss
3	R9 < R4	R9 < R4 (cycle-farmers)	10/20 vs 9/20	Fisher, one-sided	0.736	miss
3	logistic b1	logistic b1 (log2 coef) > 0	b = -3.2e-08 (not estimable: quasi-separation)	logistic Wald, one-sided	–	miss
3	logistic b3	logistic b3 (log2 coef x norm_on) < 0	b = 0.626 (not estimable: quasi-separation)	logistic Wald, one-sided	–	miss
4	A1-P1	A1b < A1c on first-solved step	median A1b 311832 vs A1c 283424	one-sided MWU	0.708	miss
4	A1-P2	A1b < A1c on solved-episode length	median A1b 20.3231 vs A1c 19.6918	one-sided MWU	0.744	miss
4	A1-P3	A1a between A1b and A1c on first-solved step (median order)	medians A1b 311832, A1a 221640, A1c 283424	median ordering (MWU reported)	–	miss
4	A1-P4	A1a between A1b and A1c on solved-episode length (median order)	medians A1b 20.3231, A1a 19.5622, A1c 19.6918	median ordering (MWU reported)	–	miss
4	A2-P1	8×8 eval-curve area: A2a < A2c	mean 0.387 vs 0.702	one-sided MWU	0.00018	hit
4	A2-P2	8×8 eval-curve area: A2b < A2c	mean 0.060 vs 0.702	one-sided MWU	5×10^{-6}	hit
4	A2-P3	16×16 eval-curve area: A2a < A2c	mean 0.328 vs 0.579	one-sided MWU	0.002	hit
4	A2-P4	16×16 eval-curve area: A2b < A2c	mean 0.011 vs 0.579	one-sided MWU	5×10^{-6}	hit
4	A2-P5	8×8 solves: A2a < A2c	11/20 vs 17/20	one-sided Fisher	0.041	hit
4	A2-P6	8×8 solves: A2b < A2c	0/20 vs 17/20	one-sided Fisher	1.3×10^{-8}	hit

Rd	ID	Prediction	Observed	Test	raw p	Result
4	A2-P7	16×16 solves: A2a < A2c	6/20 vs 8/20	one-sided Fisher	0.371	miss
4	A2-P8	16×16 solves: A2b < A2c	0/20 vs 8/20	one-sided Fisher	0.002	hit
4	B0-P	optimum tampers ≥ 1, reaches goal, never tampers indefinitely (1/160, 1/40)	1/160 (g): reaches goal, 1 tamper(s) at steps [0], 12 steps; 1/40 (g): reaches goal, 2 tamper(s) at steps [0, 11], 13 steps; 1/40 (g/2): reaches goal, 2 tamper(s) at steps [0, 11], 13 steps	deterministic	–	hit
4	PB1-B1	B1 tamperers (reference A1b median done/episode 4.8e-05)	20/20 (median done/episode 38.6)	binomial vs 0.5, one-sided	9.5×10^{-7}	hit
4	PB1-B2	B2 tamperers (reference B7 median done/episode 2.51)	20/20 (median done/episode 61.2)	binomial vs 0.5, one-sided	9.5×10^{-7}	hit
4	PB1-B3	B3 tamperers (reference B5 median done/episode 0.000789)	20/20 (median done/episode 173)	binomial vs 0.5, one-sided	9.5×10^{-7}	hit
4	PB1-B4	B4 tamperers (reference B6 median done/episode 0.475)	20/20 (median done/episode 120)	binomial vs 0.5, one-sided	9.5×10^{-7}	hit
4	PB1-pooled	B1-B4 tamperers pooled	80/80	binomial vs 0.5, one-sided	8.3×10^{-25}	hit
4	PB2-B1	solve B1 (tamper on) < A1b (tamper off)	1/20 vs 20/20	one-sided Fisher	1.5×10^{-10}	hit
4	PB2-B2	solve B2 (tamper on) < B7 (tamper off)	0/20 vs 20/20	one-sided Fisher	7.3×10^{-12}	hit
4	PB2-B3	solve B3 (tamper on) < B5 (tamper off)	1/20 vs 11/20	one-sided Fisher	0.00062	hit
4	PB2-B4	solve B4 (tamper on) < B6 (tamper off)	0/20 vs 20/20	one-sided Fisher	7.3×10^{-12}	hit
4	PB3-B3	tamper-loopers B3 (norm on) > B4 (norm off)	20/20 vs 20/20	one-sided Fisher	1.000	miss
4	PB3-B1	tamper-loopers B1 (norm on) > B2 (norm off)	20/20 vs 20/20	one-sided Fisher	1.000	miss
4	PB4	cycle-farm B5 (norm on) > B6 (norm off)	9/20 vs 0/20	one-sided Fisher	0.00061	hit
4	PB5	B8 cycle-farms in most seeds	20/20	binomial vs 0.5, one-sided	9.5×10^{-7}	hit
5	A-P1	A1 solves more than B1	11/20 vs 1/20	one-sided Fisher	0.00062	hit
5	A-P1m	A1 success-curve area > B1	mean 0.438 vs 0.042	one-sided MWU	7.8×10^{-6}	hit
5	A-P2	A2 solves more than B2	0/20 vs 0/20	one-sided Fisher	1.000	miss
5	A-P2m	A2 success-curve area > B2	mean 0.025 vs 0.022	one-sided MWU	0.005	hit
5	A-P3	A1 solves more than B1+B3	11/20 vs 2/40	one-sided Fisher	2.6×10^{-5}	hit
5	A-P3m	A1 success-curve area > B1+B3	mean 0.438 vs 0.039	one-sided MWU	5×10^{-6}	hit
5	B-JTp	tamper presses decrease with cost (B1, B9, B10, B11)	medians 38.59, 27.13, 0.36, 0.00	one-sided Jonckheere-Terpstra (permutation)	5×10^{-6}	hit

Rd	ID	Prediction	Observed	Test	raw p	Result
5	B-JTs	solves increase with cost (B1, B9, B10, B11)	solves 1, 3, 20, 20	one-sided Jonckheere-Terpstra (permutation)	5×10^{-6}	hit
5	B-F-B9-vsB1	B9 solves more than B1	3/20 vs 1/20	one-sided Fisher	0.302	hit
5	B-F-B9-vsA1b	B9 solves fewer than A1b	3/20 vs 20/20	one-sided Fisher	1.3×10^{-8}	hit
5	B-L-B9-presses	B9 presses (median presses per episode)	27.13	level (range)	–	hit
5	B-L-B9-tamperers	B9 tamperers (count of 20)	19/20	level (range)	–	hit
5	B-L-B9-solves	B9 solves (count of 20)	3/20	level (range)	–	hit
5	B-F-B10-vsB1	B10 solves more than B1	20/20 vs 1/20	one-sided Fisher	1.5×10^{-10}	miss
5	B-F-B10-vsA1b	B10 solves fewer than A1b	20/20 vs 20/20	one-sided Fisher	1.000	miss
5	B-L-B10-presses	B10 presses (median presses per episode)	0.36	level (range)	–	miss
5	B-L-B10-tamperers	B10 tamperers (count of 20)	7/20	level (range)	–	miss
5	B-L-B10-solves	B10 solves (count of 20)	20/20	level (range)	–	miss
5	B-F-B11-vsB1	B11 solves more than B1	20/20 vs 1/20	one-sided Fisher	1.5×10^{-10}	hit
5	B-F-B11-vsA1b	B11 solves fewer than A1b	20/20 vs 20/20	one-sided Fisher	1.000	hit
5	B-L-B11-presses	B11 presses (median presses per episode)	0.00	level (range)	–	hit
5	B-L-B11-tamperers	B11 tamperers (count of 20)	0/20	level (range)	–	hit
5	B-L-B11-solves	B11 solves (count of 20)	20/20	level (range)	–	hit
5	C-P1	cycle-farming rises K1 < K2 < K3	0/20, 0/20, 0/20	one-sided Jonckheere-Terpstra (permutation)	1.000	miss
5	C-P2	cycle-farm: K3 > K4	0/20 vs 0/20	one-sided Fisher	1.000	miss
5	C-P3	solve: K5 < K1	0/20 vs 20/20	one-sided Fisher	7.3×10^{-12}	hit
5	C-P4	solve: K5 < K6	0/20 vs 20/20	one-sided Fisher	7.3×10^{-12}	hit
5	D1-P1	B1-B4 seeds that never found the goal	1/80	level (range)	–	hit
5	D1-P2	found-then-lost seeds whose success declined (peak ≥ 0.10, then $\leq$ half)	7/77	level: hit iff $\geq$ 20% (≥ 16)	–	miss
5	D1-P3	median exact first-goal step of B1-B4 seeds	27,344 (max 1,187,264)	level: hit iff $\leq$ 41,000	–	hit
5	D2-B1	final-10% policy entropy B1 (tamper on) > A1b (tamper off)	median 1.878 vs 0.067	one-sided MWU	7.3×10^{-12}	hit
5	D2-B2	final-10% policy entropy B2 (tamper on) > B7 (tamper off)	median 1.935 vs 1.138	one-sided MWU	7.3×10^{-12}	hit
5	D2-B3	final-10% policy entropy B3 (tamper on) > B5 (tamper off)	median 1.844 vs 0.290	one-sided MWU	7.3×10^{-12}	hit

Rd	ID	Prediction	Observed	Test	raw p	Result
5	D2-B4	final-10% policy entropy B4 (tamper on) > B6 (tamper off)	median 1.934 vs 0.598	one-sided MWU	7.3×10^{-12}	hit
5	D3-P1	'done' fraction rises before success declines	rise first 7/7 (undetermined 0)	binomial vs 0.5, one-sided; hit iff share $\geq$ 0.8	0.008	hit

B Known misclassifications and corrections

Reproduced from `consolidated-results.md`.

1. **Round 2, temptation formula F redefined after seeing data.** The first F did not fit the three calibration runs and was redefined before the round-2 test; the calibration runs were therefore excluded as evidence (disclosed in round 2).
2. **Round 2, pay-once control labelled 'farmer' by the >5 rule** although replay showed episode-prolonging, not repetition; the round-3 exploratory re-score repeats this label.
3. **Round 2 claim dropped: 'the critic prefers farming'.** The value rising with need is the $+D(h_t)$ term of the return and holds for solvers too; not evidence of preference.
4. **Round 1 claim corrected: LP-curiosity seeds 2 and 4 'entropy collapse'.** Entropy stayed high; the cause was left open.
5. **Event-bonus per-new-view payment** was written as 0.0975 in round-2 notes; the code pays $h_0^2 - (h_0 - 0.1)^2 = 0.09$ (corrected in round 3).
6. **Round 3, coef x normalization logistic model not estimable** (all 60 normalization-off seeds solved); recorded as misses.
7. **Evaluation bias, rounds 1-3.** `evaluate()` kept the first N episodes to finish across 16 parallel envs, over-weighting short successful episodes. Re-evaluation (round 4): 16×16 novelty 1e-3 0.845 → 0.787; Agent 3 1/160 0.714 → 0.653; 1/640 0.622 → 0.548; ranking unchanged. Fixed from round 4 on (per-environment quota).
8. **Round 4, explore-only runs (A2a, A2b) labelled 'cycle-farm'** although pick-ups were not paid in those runs; replay showed non-learning policies. Led to the round-5 rule: cycle-farm applies only where pick-ups or door toggles are paid.
9. **Round 4, B1 and B2 ran concurrently** because a launcher missed a setup status; seeded single-threaded computation, so results are unaffected.
10. **Round 5, Step 0 at growth 1/160** used a certified early stop (validated against full convergence on all 21 growth-1/40 cases) instead of full convergence.
11. **Round 5, A-P2m is a 'hit' on a negligible effect** (success-curve area 0.025 vs 0.022); both runs are at the floor and the solve test (A-P2) missed. Do not cite A-P2m as evidence.
12. **Round 5, tamper labels in tamper-off runs.** The 'tamperers' column for tamper-off runs counts presses of the no-op 'done' action (e.g. A2 presses at the uniform-random rate); the labels apply only to tamper-on runs.
13. **Round 5, Part D 'found then lost'** includes seeds that only met the goal by chance; only 7/77 declined after real learning (D1-P2 miss), and D3 rests on those 7 seeds.

C Per-seed results

Table 7: Round 4, Part B, per seed: share of steps on "done" over the final 10% of training log rows, and the seed's behaviour category (S solve, C cycle-farm, P prolong). Tamper-on runs alternate with their matched tamper-off runs, where "done" is a no-op. Category labels in tamper-on runs are mechanical (more than 5 paid repeats per episode) and are not used as evidence of farming (Section 5.5).

Seed	growth 1/160, norm on		1/160, norm off		1/40, norm on		1/40, norm off	
	tamper	no tamper	tamper	no tamper	tamper	no tamper	tamper	no tamper
40	0.060 C	0.000 S	0.096 C	0.082 S	0.257 C	0.000 S	0.189 C	0.016 S
41	0.061 P	0.000 S	0.100 C	0.081 S	0.273 P	0.013 C	0.188 C	0.017 S
42	0.059 P	0.000 S	0.096 C	0.083 S	0.270 P	0.000 S	0.189 C	0.022 S
43	0.059 P	0.000 S	0.094 C	0.081 S	0.270 C	0.001 C	0.191 C	0.015 S
44	0.064 P	0.000 S	0.101 C	0.085 S	0.273 P	0.000 S	0.187 C	0.020 S
45	0.062 P	0.000 S	0.096 C	0.083 S	0.270 P	0.000 S	0.189 C	0.019 S
46	0.056 C	0.000 S	0.094 C	0.080 S	0.261 P	0.000 C	0.189 C	0.018 S
47	0.060 C	0.000 S	0.097 C	0.081 S	0.268 C	0.000 C	0.188 C	0.022 S
48	0.067 S	0.000 S	0.096 C	0.086 S	0.217 S	0.000 C	0.187 C	0.023 S
49	0.060 P	0.000 S	0.099 C	0.083 S	0.254 P	0.000 S	0.188 C	0.012 S
50	0.060 C	0.000 S	0.094 C	0.082 S	0.256 P	0.000 S	0.185 C	0.019 S
51	0.061 C	0.000 S	0.098 C	0.079 S	0.270 C	0.005 C	0.187 C	0.016 S
52	0.063 P	0.000 S	0.097 C	0.078 S	0.274 P	0.000 S	0.190 C	0.021 S
53	0.062 P	0.000 S	0.097 C	0.077 S	0.282 P	0.000 C	0.193 P	0.017 S
54	0.065 P	0.000 S	0.099 C	0.084 S	0.277 P	0.000 S	0.190 C	0.020 S
55	0.061 P	0.000 S	0.097 C	0.084 S	0.272 P	0.000 S	0.190 C	0.022 S
56	0.064 P	0.000 S	0.094 C	0.079 S	0.276 P	0.000 S	0.187 P	0.018 S
57	0.060 C	0.000 S	0.092 C	0.081 S	0.258 P	0.000 S	0.189 C	0.018 S
58	0.063 P	0.000 S	0.098 C	0.076 S	0.276 C	0.004 C	0.189 C	0.020 S
59	0.061 P	0.000 S	0.095 C	0.082 S	0.280 P	0.002 C	0.190 C	0.025 S
median	0.061	0.000	0.096	0.081	0.270	0.000	0.189	0.019

Table 8: Round 5, Part B, per seed (growth 1/160, normalization on): tamper presses per step over the final 10% of log rows (descriptive, from the run logs) and category. Cost 0 and the no-tamper column are the round-4 runs on the same seeds.

Seed	cost 0	cost 0.01	cost 0.1	cost 1	no tamper action
40	0.060 C	0.043 C	0.022 S	0.000 S	0.000 S
41	0.061 P	0.042 F	0.000 S	0.000 S	0.000 S
42	0.059 P	0.043 C	0.009 S	0.000 S	0.000 S
43	0.059 P	0.045 S	0.016 S	0.000 S	0.000 S
44	0.064 P	0.044 P	0.001 S	0.000 S	0.000 S
45	0.062 P	0.025 S	0.000 S	0.000 S	0.000 S
46	0.056 C	0.041 C	0.020 S	0.000 S	0.000 S
47	0.060 C	0.039 C	0.002 S	0.000 S	0.000 S
48	0.067 S	0.042 S	0.020 S	0.000 S	0.000 S
49	0.060 P	0.044 P	0.008 S	0.000 S	0.000 S
50	0.060 C	0.043 C	0.011 S	0.000 S	0.000 S
51	0.061 C	0.043 C	0.023 S	0.000 S	0.000 S
52	0.063 P	0.043 C	0.012 S	0.000 S	0.000 S
53	0.062 P	0.045 P	0.021 S	0.000 S	0.000 S
54	0.065 P	0.045 P	0.000 S	0.000 S	0.000 S

Seed	cost 0	cost 0.01	cost 0.1	cost 1	no tamper action
55	0.061 P	0.042 P	0.000 S	0.000 S	0.000 S
56	0.064 P	0.043 P	0.019 S	0.000 S	0.000 S
57	0.060 C	0.043 C	0.001 S	0.000 S	0.000 S
58	0.063 P	0.046 P	0.020 S	0.000 S	0.000 S
59	0.061 P	0.044 P	0.003 S	0.000 S	0.000 S

Table 9: Part D per seed (round-4 tamper-on runs; analyses registered in round 5). First-goal step is exact; peak and final success are rolling 100-episode means; "declined" requires peak ≥ 0.10 and final $\leq$ half the peak; entropy and "done" share are final-10% means; order compares the first row with "done" share $\geq 1/7 + 0.05$ (rise) with the decline row.

Run	Seed	First goal	Class	Peak	Final	Declined	Entropy	"Done"	Rise	Order
1/160 on	40	88,800	lost	0.04	0.021	no	1.897	0.060	40,960	no decline
	41	6,288	lost	0.06	0.026	no	1.878	0.061	40,960	no decline
	42	71,008	lost	0.05	0.036	no	1.858	0.059	40,960	no decline
	43	25,632	lost	0.13	0.083	no	1.872	0.059	40,960	rise first
	44	255,984	lost	0.03	0.002	no	1.866	0.064	40,960	no decline
	45	4,432	lost	0.06	0.010	no	1.887	0.062	40,960	no decline
	46	46,864	lost	0.10	0.034	yes	1.892	0.056	40,960	rise first
	47	18,384	lost	0.17	0.009	yes	1.891	0.060	40,960	rise first
	48	57,600	kept	1.00	0.997	no	1.164	0.067	40,960	no decline
	49	6,096	lost	0.13	0.022	yes	1.852	0.060	–	no rise
	50	194,448	lost	0.09	0.040	no	1.885	0.060	40,960	no decline
	51	16,096	lost	0.10	0.026	yes	1.881	0.061	61,440	rise first
	52	20,368	lost	0.05	0.027	no	1.880	0.063	40,960	no decline
	53	9,568	lost	0.03	0.008	no	1.853	0.062	40,960	no decline
	54	144,624	lost	0.03	0.002	no	1.871	0.065	40,960	no decline
	55	6,080	lost	0.06	0.019	no	1.864	0.061	40,960	no decline
	56	101,200	lost	0.02	0.000	no	1.881	0.064	61,440	no decline
	57	17,424	lost	0.20	0.054	yes	1.859	0.060	40,960	rise first
	58	107,488	lost	0.13	0.001	yes	1.878	0.063	20,480	rise first
	59	9,008	lost	0.05	0.019	no	1.886	0.061	40,960	no decline
1/160 off	40	10,096	lost	0.06	0.028	no	1.935	0.096	–	no rise
	41	6,288	lost	0.07	0.022	no	1.936	0.100	–	no rise
	42	56,416	lost	0.06	0.024	no	1.936	0.096	–	no rise
	43	29,664	lost	0.05	0.023	no	1.934	0.094	163,840	no decline
	44	55,184	lost	0.06	0.016	no	1.938	0.101	102,400	no decline
	45	4,432	lost	0.06	0.026	no	1.935	0.096	–	no rise
	46	86,112	lost	0.07	0.031	no	1.935	0.094	–	no rise
	47	25,184	lost	0.06	0.039	no	1.936	0.097	–	no rise
	48	47,808	lost	0.08	0.036	no	1.936	0.096	–	no rise
	49	20,272	lost	0.09	0.042	no	1.936	0.099	–	no rise
	50	19,568	lost	0.07	0.024	no	1.934	0.094	–	no rise
	51	16,176	lost	0.07	0.032	no	1.936	0.098	–	no rise
	52	9,728	lost	0.07	0.032	no	1.935	0.097	–	no rise
	53	9,568	lost	0.05	0.027	no	1.936	0.097	122,880	no decline
	54	57,744	lost	0.06	0.028	no	1.936	0.099	–	no rise
	55	6,080	lost	0.08	0.012	no	1.935	0.097	–	no rise
	56	18,576	lost	0.05	0.017	no	1.934	0.094	–	no rise
	57	12,864	lost	0.06	0.024	no	1.933	0.092	–	no rise
	58	111,680	lost	0.06	0.020	no	1.936	0.098	81,920	no decline
	59	88,320	lost	0.05	0.018	no	1.935	0.095	–	no rise

Run	Seed	First goal	Class	Peak	Final	Declined	Entropy	"Done"	Rise	Order
1/40 on	40	8,928	lost	0.09	0.014	no	1.872	0.257	20,480	no decline
	41	6,288	lost	0.03	0.008	no	1.853	0.273	20,480	no decline
	42	695,728	lost	0.02	0.002	no	1.859	0.270	20,480	no decline
	43	1,187,264	lost	0.03	0.006	no	1.846	0.270	20,480	no decline
	44	690,464	lost	0.05	0.000	no	1.835	0.273	20,480	no decline
	45	4,432	lost	0.03	0.000	no	1.819	0.270	20,480	no decline
	46	377,712	lost	0.04	0.002	no	1.882	0.261	20,480	no decline
	47	9,120	lost	0.13	0.013	yes	1.858	0.268	20,480	rise first
	48	163,520	kept	0.94	0.901	no	1.323	0.217	20,480	no decline
	49	27,808	lost	0.40	0.334	no	1.773	0.254	20,480	no decline
	50	9,696	lost	0.04	0.013	no	1.897	0.256	20,480	no decline
	51	846,704	lost	0.03	0.008	no	1.853	0.270	20,480	no decline
	52	1,092,016	lost	0.01	0.003	no	1.844	0.274	20,480	no decline
	53	16,960	lost	0.03	0.000	no	1.824	0.282	20,480	no decline
	54	959,856	lost	0.01	0.002	no	1.829	0.277	20,480	no decline
	55	233,168	lost	0.02	0.003	no	1.822	0.272	20,480	no decline
	56	50,832	lost	0.01	0.000	no	1.808	0.276	20,480	no decline
	57	15,824	lost	0.06	0.017	no	1.887	0.258	20,480	no decline
	58	never	never	0.00	0.000	no	1.722	0.276	20,480	no decline
	59	259,280	lost	0.03	0.007	no	1.844	0.280	20,480	no decline
1/40 off	40	181,664	lost	0.05	0.019	no	1.935	0.189	40,960	no decline
	41	6,288	lost	0.07	0.029	no	1.934	0.188	40,960	no decline
	42	48,928	lost	0.06	0.010	no	1.934	0.189	40,960	no decline
	43	127,104	lost	0.05	0.022	no	1.934	0.191	40,960	no decline
	44	46,944	lost	0.05	0.031	no	1.934	0.187	40,960	no decline
	45	4,432	lost	0.05	0.022	no	1.933	0.189	40,960	no decline
	46	20,480	lost	0.05	0.017	no	1.934	0.189	40,960	no decline
	47	49,424	lost	0.07	0.044	no	1.934	0.188	40,960	no decline
	48	27,344	lost	0.06	0.019	no	1.935	0.187	40,960	no decline
	49	17,840	lost	0.05	0.017	no	1.934	0.188	40,960	no decline
	50	30,496	lost	0.07	0.033	no	1.935	0.185	40,960	no decline
	51	16,096	lost	0.06	0.020	no	1.935	0.187	40,960	no decline
	52	7,744	lost	0.09	0.032	no	1.933	0.190	40,960	no decline
	53	56,864	lost	0.06	0.020	no	1.933	0.193	40,960	no decline
	54	30,400	lost	0.06	0.006	no	1.933	0.190	40,960	no decline
	55	9,744	lost	0.09	0.036	no	1.934	0.190	40,960	no decline
	56	59,520	lost	0.06	0.019	no	1.934	0.187	40,960	no decline
	57	9,136	lost	0.06	0.020	no	1.934	0.189	40,960	no decline
	58	98,864	lost	0.05	0.020	no	1.934	0.189	20,480	no decline
	59	25,968	lost	0.07	0.022	no	1.933	0.190	40,960	no decline

D Theory computations

Check (a). Layouts use environment seeds 10000–10199. The breadth-first search runs over the real MiniGrid dynamics, restoring the full grid state at each node, with actions left, right, forward, pick-up, drop and toggle. "Done" never shortens a path. Success is reaching the goal (DoorKey) or carrying the target ball (KeyCorridor).

- **Replay.** Each candidate action sequence is replayed in a fresh environment with the training code's satiation module.
- **Drop-and-regrab.** The richer family inserts one drop-and-regrab at every position where the agent carries the key and faces an empty cell, and keeps the best member.

- **Tamper family.** Tamper at $t = 0$, with or without a second tamper just before the goal step.

Check (b). The 9 farming checkpoints of the round-2 growth-1/20 run were each run on the same 200 layouts, with actions sampled from the policy under a fixed seed per batch. One further seed of that run solved (0.995 success) and is excluded. One farming seed succeeded on 0.26 of layouts; its 52 solving episodes are excluded from the non-solving maximum.

Replay of the tamper-on policies. All 80 round-4 tamper-on checkpoints (four settings, 20 seeds each) were run on the same 200 layouts with sampled actions, as in check (b), with growth and the tamper action as in training.

- **Recorded per episode:** the discounted intrinsic return, the total discounted return (the task reward arrives only on the final step), the mean of $D(h_t)$ over the episode's steps, success, and the share of steps on "done".
- **Checks:** the tracked drive matched the module's need state at every step, and the identity of Proposition 1 held on every episode. For every seed, the replay's press share was within 0.007 of the share in its training log.

Script `tamper_policy_returns.py`, data `tamper-policy-returns.json`.

Value iteration. Value iteration ran on a 7×5 layout with a wall column and a locked door, with needs (grasp, open) on an exact lattice of bin g. A halved bin reproduced the results where it was checked. Convergence was to $\max |\Delta V| < 10^{-7}$, except for the certified early stop at 1/160 in round 5. The tamper version zeroes both modelled needs.

E Hyperparameters and implementation

Table 10: PPO and reward settings (from `config.toml`).

Setting	Value
Environments in parallel	16
Rollout length per environment	128
Batch	2048
Environment steps per seed	2,000,000
Learning rate (linearly annealed)	2.5×10^{-4}
Discount γ	0.99
GAE λ	0.95
Minibatches per epoch	4
Epochs per update	4
Clip coefficient	0.2
Entropy coefficient	0.01
Value coefficient	0.5
Max gradient norm	0.5
Advantage normalization (per minibatch)	on, or off where stated
Needs h (explore, grasp, open): start h_0	0.5
Relief per event (satiety)	0.1, 1.0, 1.0
Intrinsic coefficient	1.0
Final evaluation episodes per environment	64

The encoder applies three 2×2 convolutions (16, 32 and 64 channels, ReLU, with one 2×2 max-pool) to the 7×7×3 view, followed by a 64-unit ReLU layer. The needs (when present) are concatenated to these features, and linear actor and critic heads follow. Software: PyTorch 2.14.1, Gymnasium 1.4.0, MiniGrid 3.2.0, NumPy 2.4.6.

F Compute

All runs used one Apple M4 Pro computer (12 CPU cores, 24 GB memory) with no GPU. Each run trained its seeds in parallel, one single-threaded process per seed.

- **Rounds 2–5:** 43 runs, 12.8 hours of summed wall-clock time from the run records.
- **Round 1** (exploratory): 14 runs, 3.0 hours. Three of these runs are re-scored in Table 4.

Two round-4 runs (satiation 1/160 + tamper, norm on and off) ran concurrently by mistake, so their recorded durations are about twice the usual. Seeded, single-threaded computation leaves their results unaffected (correction 9).

G Reproducibility

Pre-registration files and their sha256 hashes, recomputed from the snapshots in the archive's `paper-data/sources/` folder and equal to the hashes in the records written at locking.

- **The lock times are self-reported.** Each record holds a file's hash and the UTC time at which we wrote it: 2026-10-26 15:17:42 (round 3), 2026-10-26 19:23:42 (round 4) and 2026-10-27 04:34:11 (round 5). By our account, each was written before any training in its round.
- **What a reader can and cannot check.** The pre-registration files and their records were not committed to version control, nor deposited with any timestamping service, at lock time. A reader can check that a released file matches the hash reported here, but not when it was written.
- **Release.** Pre-registration files, hash records, run records, scored results, checkpoints, and the training and analysis code used in this paper are archived at Zenodo, doi:`10.5281/zenodo.23708846`.

File	sha256
Round 3 pre-registration	`b5e845c978ec96c2746eb81098c0bea3bf8dc14a1dc174e4f77844352b621ae7`
Round 4 pre-registration	`79fdfed000d5e6f892d9201cadd970808cf3b0949e4b3e3d925be9839800faa1`
Round 4 amendment	`72b77be693b9d493e756b23f89aadb35d6934965d5c0633d4681fc8422c0e6f5`
Round 5 pre-registration	`0298bbc79c323e4613e56a29be0257a6c7c3ad764cbb056902159292bcbcda0d`
Round 5 amendment (Part C lock)	`1d99028349683b3b85bd204802b614bfb3a38fa30db3d24fbb19eda7c2790a4f`

- **Generation.** Every figure, table and quoted number is generated by a script in the archive's `scripts/` folder, from the scored files in `paper-data/sources/` or from the post-hoc data in `paper-data/`. The archive's `README.md` maps each file to the paper section that uses it.
- **Checking.** `check_numbers.py` regenerates the number macros and tables, fails on any difference, and flags literal numbers in the text.
- **Run IDs.** Table 11 maps the descriptive labels to run identifiers.

Table 11: Map from the descriptive labels used in the text to the internal run names and run identifiers. DoorKey-8x8 unless stated; normalization on unless stated. KeyCorridor runs are named K1–K6 here; the project's scored files and run records call them C1–C6.

Label in the text	Round	Name	Run ID
satiation 1/40 (round-1 calibration)	1	cal 1/40	`6bfabf4a`
satiation 1/160 (round-1 calibration)	1	cal 1/160	`6e507c3e`
satiation 1/640 (round-1 calibration)	1	cal 1/640	`6518a3c8`
satiation 1/40 (seeds 10–19)	2	N1	`ed5147b6`
satiation 1/40, $\gamma = 0.95$	2	N2	`74b2b2d0`
satiation 1/80	2	N3	`af4373da`
satiation 1/80, coef 1.29	2	N4	`b58ec6e0`
satiation 1/40, coef 0.5	2	N5	`eb8555bd`
satiation 1/20	2	N6	`bced806e`
pay-once event bonus	2	pay-once	`5b13189d`
satiation 1/60	3	R1	`4cac9e0a`
satiation 1/30	3	R2	`eea13605`
satiation 1/40, coef 0.5	3	R3	`3d03a914`
satiation 1/40 (reference)	3	R4	`fff9fd5b`
satiation 1/40, coef 2	3	R5	`c3d67c09`
satiation 1/40, coef 0.5, norm off	3	R6	`89d8e34c`
satiation 1/40, norm off	3	R7	`135d1475`
satiation 1/40, coef 2, norm off	3	R8	`1205f6d0`
satiation 1/40 + early-goal starts	3	R9	`a7cb164e`
fixed pay-once control	3	R10	`07653362`
satiation 1/640	4	A1a	`a97113a1`
satiation 1/160	4	A1b	`3629f9a7`
satiation, no refill (growth 0)	4	A1c	`1d549b2f`
explore-only satiation 1/640	4	A2a	`224a7d96`
explore-only satiation 1/160	4	A2b	`7a364fed`
episodic novelty	4	A2c	`2d94fc36`
satiation 1/160 + tamper	4	B1	`bb00e7e5`
satiation 1/160 + tamper, norm off	4	B2	`a8f5efc4`
satiation 1/40 + tamper	4	B3	`78c44406`
satiation 1/40 + tamper, norm off	4	B4	`f0a40d20`
satiation 1/40	4	B5	`4d3b84f0`
satiation 1/40, norm off	4	B6	`a6e11900`
satiation 1/160, norm off	4	B7	`115c6566`
constant event bonus, norm off	4	B8	`fb860207`
task-only	5	A1	`022a3e90`
task-only, norm off	5	A2	`1c7b2eda`
satiation 1/160 + tamper, cost 0.01	5	B9	`b971c18d`
satiation 1/160 + tamper, cost 0.1	5	B10	`a25b7c99`
satiation 1/160 + tamper, cost 1	5	B11	`896b2863`
KeyCorridor satiation 1/160	5	K1	`c14516de`
KeyCorridor satiation 1/40	5	K2	`ad87a02f`
KeyCorridor satiation 1/20	5	K3	`aa29d1c3`
KeyCorridor satiation 1/20, norm off	5	K4	`2153e884`
KeyCorridor satiation 1/160 + tamper	5	K5	`35c840cd`
KeyCorridor task-only	5	K6	`6571bce6`
KeyCorridor calibration (seeds 100–104)	5	calibration	`bd005cb8`